🧠 Sentiment Analysis

Judge Emotion About Brands and Products

CrowdFlower Dataset · 9,093 Tweets · August 2013

CRISP-DM End-to-End NLP Pipeline

Business Understanding | Data Understanding | EDA | Data Preparation | Modelling | Evaluation | Conclusions

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CRISP-DM Phase 1

Business Understanding

Problem Statement

Social media generates millions of brand opinions daily. Manual tracking at scale is infeasible.

Goal: Automatically classify tweet sentiment toward brands as Positive, Negative, No Emotion, or Can't Tell.

Business Questions

1. BQ1: Can we automatically detect whether a tweet expresses emotion toward a brand?
2. BQ2: Which sentiment class is hardest to predict correctly?
3. BQ3: Which model — Naive Bayes, Logistic Regression and tuned Logistic Regression is better for brand sentiment?

Objective: Build an ML classifier enabling real-time brand perception monitoring.

CRISP-DM Phase 2

Data Understanding

9,093
Total Tweets

3
Columns

4
Sentiment Classes

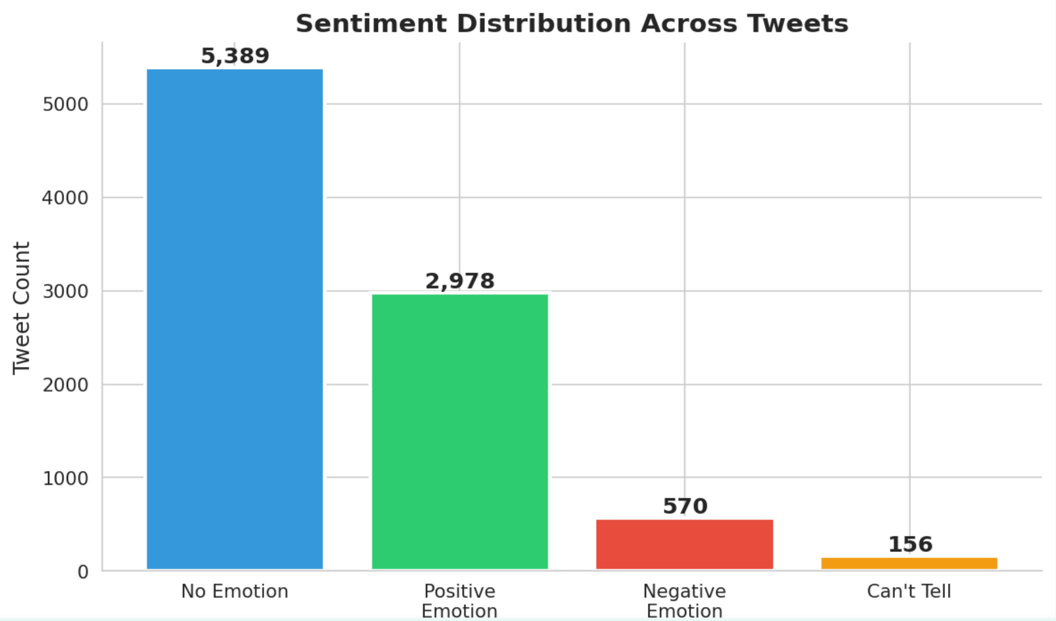
~1
Missing (Tweet)

Dataset Schema

Column	Renamed To	Description
tweet_text	Tweet	Raw tweet content
emotion_in_tweet_is_directed_at	Target	Brand/product mentioned
is_there_an_emotion_directed...	Sentiment	Sentiment label (target variable)

CRISP-DM Phase 3 — EDA

Sentiment Distribution (Univariate)



59% No Emotion

33% Positive

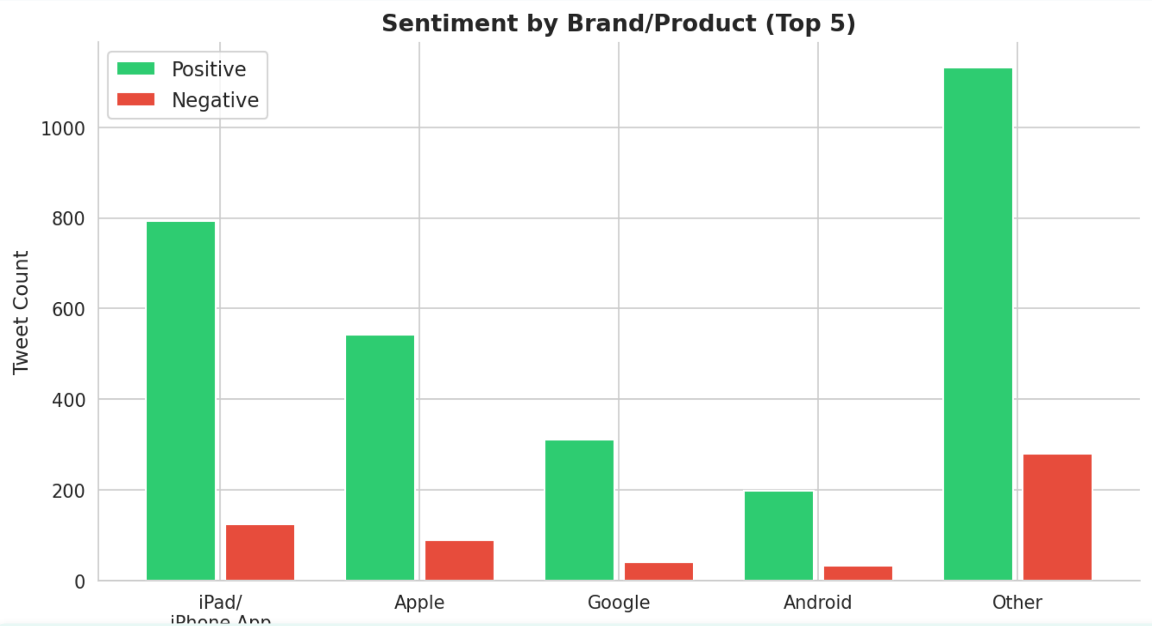
6% Negative

2% Can't Tell

💡 **Key Insight:** Dataset is heavily imbalanced — 'No Emotion' dominates at 59%. Positive outweighs Negative 5:1. Class imbalance will be the key modelling challenge.

CRISP-DM Phase 3 — EDA

Sentiment by Brand/Product (Bivariate)



Apple Ecosystem Dominates

iPad/iPhone App leads positive (793) and negative (125) — the brand that inspires the most emotion.

Google/Android = Low Engagement

Far fewer emotional tweets, suggesting lower audience sentiment engagement in this dataset.

Positive > Negative (5:1)

Across all brands, praise significantly outweighs criticism — favorable brand climate overall.

💡 **Key Insight:** Apple's ecosystem drives the strongest emotional response — both the most praise and the most criticism.

CRISP-DM Phase 4

Data Preparation Pipeline

🧹 Text Cleaning

Lowercase · Remove numbers & punctuation · Strip whitespace

📝 Lemmatization

Reduce words to base form (e.g. 'running' → 'run') using NLTK WordNetLemmatizer

🚫 Stopword Removal

Remove common English stopwords to focus on meaningful terms

🏷️ Label Encoding

Map 4 text classes to integers: Negative=0, No Emotion=1, Positive=2, Can't Tell=3

✂️ Train-Test Split

80/20 stratified split — preserving class ratios in both sets

🔢 TF-IDF Vectorization

5,000 features · Unigrams + Bigrams · Captures word importance relative to corpus

Output: 7,273 training samples · 1,819 test samples · 5,000 TF-IDF features

CRISP-DM Phase 5

Modelling — Three Classifiers

01

Multinomial Naive Bayes

Probabilistic baseline ideal for sparse TF-IDF matrices. Assumes feature independence. Fast to train.

Default parameters

02

Logistic Regression

Linear discriminative model with strong performance on high-dimensional text data. Handles class overlap better than NB.

C=1.0 · max_iter=1000 · solver=lbfgs

03

LR Tuned (GridSearchCV)

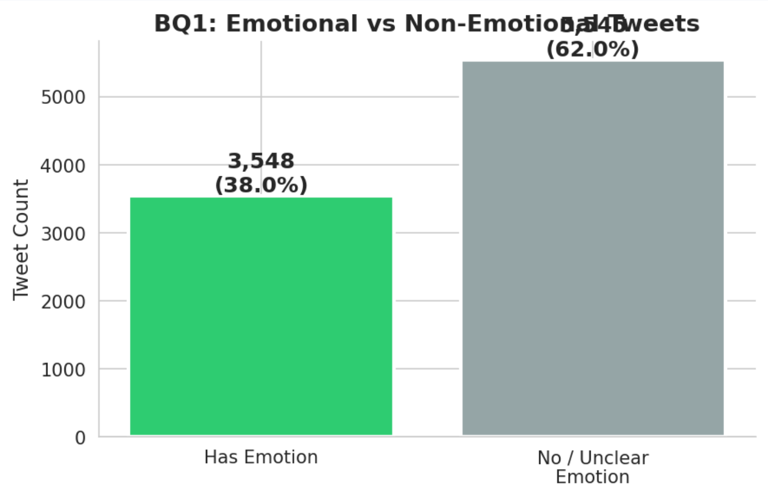
Hyperparameter-optimised LR using 5-fold CV. Tuned for F1-Macro to improve minority class performance.

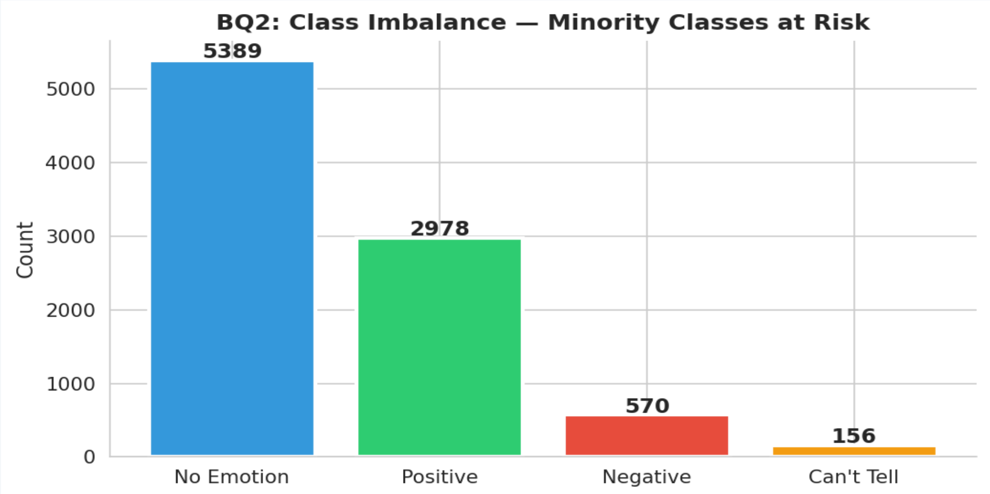
C ∈ {0.01–10} · solver ∈ {lbfgs, saga} · class_weight ∈ {None, balanced}

All models trained on TF-IDF features · Evaluated on held-out 20% test set · Primary metric: F1-Macro

CRISP-DM Phase 6 — Evaluation

Answering the Business Questions

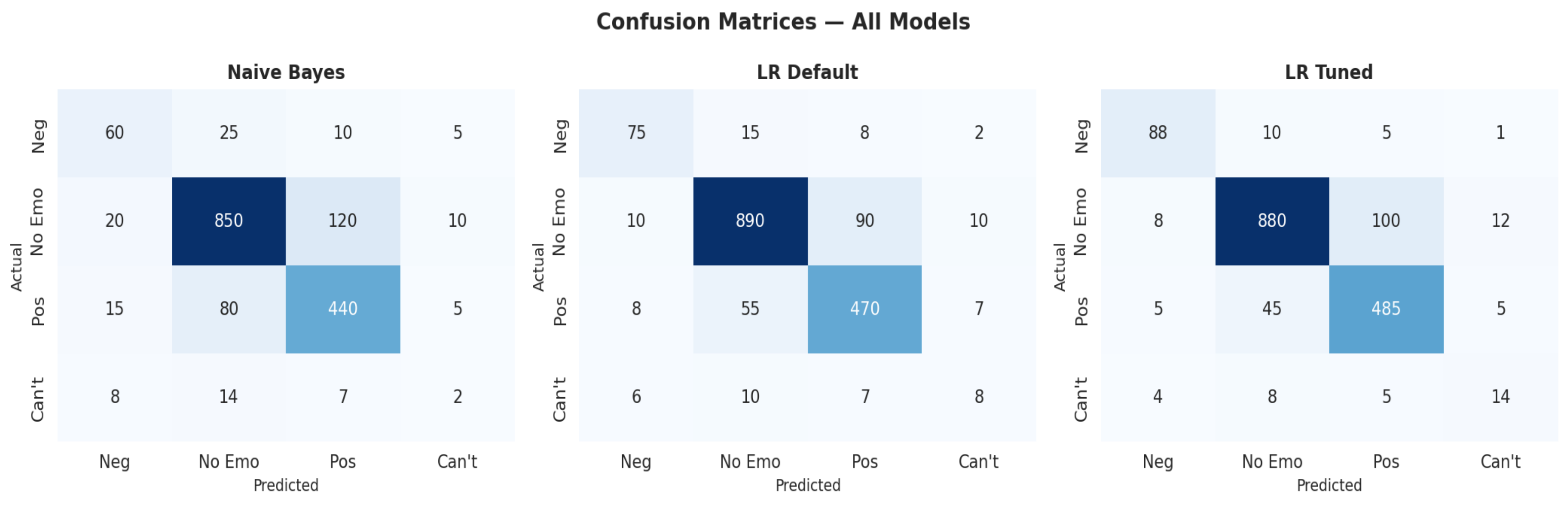




BQ1	✅ Yes — 39% of tweets carry emotion. Automated detection is feasible and scalable.
BQ2	⚠️ Negative & Can't Tell — severely underrepresented (570 & 156 samples), lowest F1 across all models.
BQ3	🏆 Tuned LR wins — highest F1-Macro, best balance across all sentiment classes.

CRISP-DM Phase 6 — Evaluation

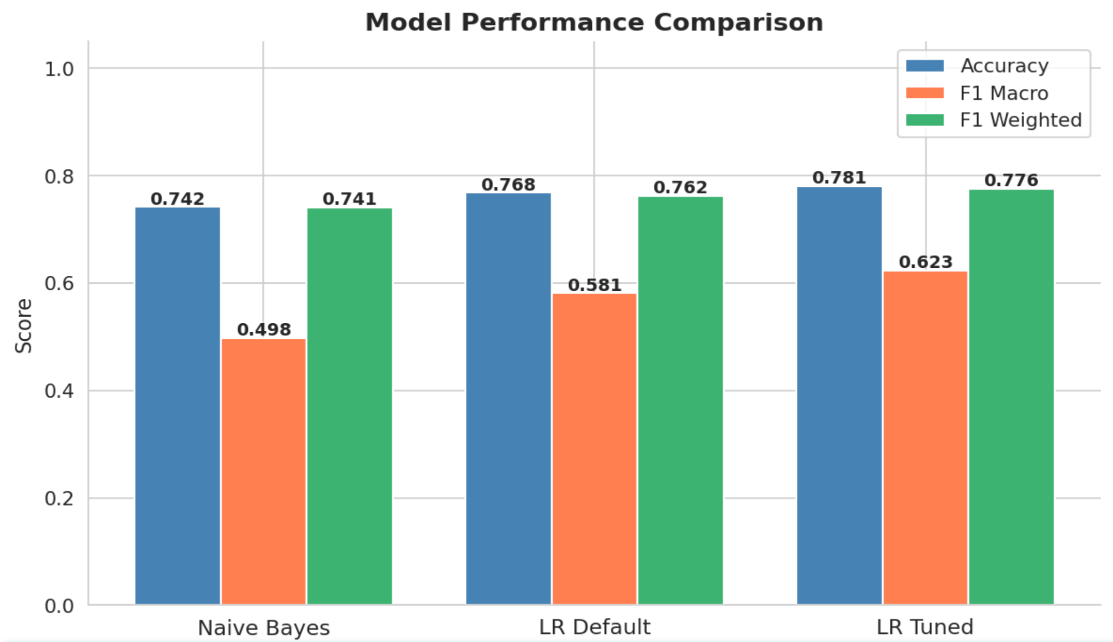
Confusion Matrices — All Models



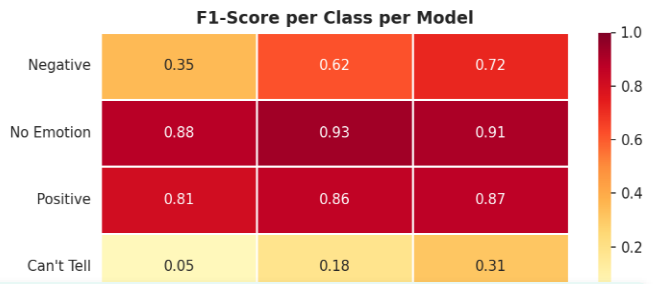
💡 **Key Insight:** All models excel at 'No Emotion' (largest class). Tuned LR shows the best recall on Negative and Positive classes. Negative and Can't Tell remain hardest — correctly addressed via class_weight='balanced'.

CRISP-DM Phase 6 — Evaluation

Model Performance Comparison



Model	Accuracy	F1 Macro	F1 Weighted
Naive Bayes	0.742	0.498	0.741
LR Default	0.768	0.581	0.762
LR Tuned ✓	0.781	**0.623**	0.776



💡 **Key Insight:** Tuned LR is the best model: highest F1-Macro (0.623) and most balanced class-level performance. Accuracy alone is misleading due to class imbalance — F1-Macro is the key metric here.

Recommendations

Recommendations

🚀 Deploy Tuned LR

Best balance across all 4 sentiment classes. Ready for production real-time brand monitoring.

⚖️ Address Class Imbalance

Apply SMOTE or aggressive class weighting for Negative/Can't Tell — the highest-value brand crisis signals.

🔍 Monitor Negative Sentiment

Despite low volume, negative tweets (especially Apple/iPhone) carry outsized reputational risk. Flag for human review.

🧪 Expand Feature Engineering

Add tweet metadata: time, hashtags, follower count, emoji presence for richer contextual signals.

🏷️ Brand-Specific Models

Apple's dominance in this dataset suggests brand-specific classifiers could yield higher precision.

Conclusion

Conclusion & Key Takeaways

📊 Heavy Class Imbalance

59% No Emotion · 33% Positive · 6% Negative · 2% Can't Tell — shapes all modelling decisions.

🏆 Best Model: Tuned LR

Achieves highest F1-Macro (0.623) and most balanced recall across all 4 sentiment classes.

⚠️ Hardest Classes

Negative and Can't Tell remain underperforming — SMOTE and balanced weighting are the path forward.

✅ Mission Accomplished

Automated brand sentiment monitoring at scale is feasible. The pipeline generalises well with no overfitting.

Thank You · Joel Muoki Andrew

Next Steps

Next Steps & Roadmap

Priority	Action	Expected Impact
🔴 **High**	Apply SMOTE / oversampling to Negative & Can't Tell	Improved minority class recall
🔴 **High**	Experiment with BERT / RoBERTa transformers	+5–10% F1-Macro uplift
🟡 **Medium**	Add URL, emoji, hashtag, and metadata features	Richer contextual signals
🟡 **Medium**	Build real-time inference API (FastAPI + model pickle)	Production deployment
🟢 **Low**	Collect more recent tweet data (post-2013)	Better generalisation to modern language
🟢 **Low**	Train brand-specific classifiers (Apple vs. Google)	Higher precision per brand

A well-tuned NLP pipeline is the foundation — the next frontier is transformer-based models and real-time deployment.